Exercise 1

Task 4. Print the number of external references (dbxrefs) and ids of the external references.

1) record.dbxrefs – external references that apply to the entire sequence record.

link the entire sequence to a project in NCBI's database, a study, or a large dataset (like BioProject or BioSample)

2) Record.features[i].qualifiers['db_xref'] – Feature-level cross-references.

external references associated with specific features, like genes, CDS (coding sequences) / protein-coding regions, etc.

Task 7. Retrieve and print all the locations of “CDS” features of the sequence (hint: check the features)

1) **Get features**
2) **Filter them by type (we need CDS)**
3) **Get location**

```
for f in record.features:
 if(f.type == "CDS"):
   print(f.type, f.location)
```

For the example gb record:

```
len(record.features)
```

```
2
```

type - the specified type of the feature
(ie. CDS, exon, repeat…)
(from library docs)

```
for f in record.features:
  print(f"{f.__dict__.keys()}")
  break
```

```
dict_keys(['location', 'type', 'id', 'qualifiers'])
```

```
for f in record.features:
  print(f.type)
```

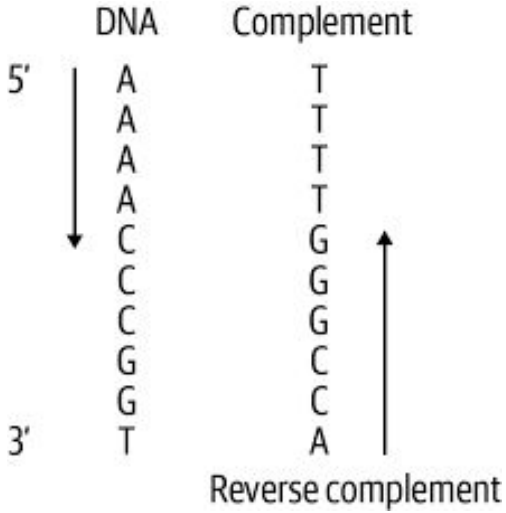
```
source
misc_feature
```

```
print("CDS features:")
for f in record.features:
  if(f.type == "CDS"):
    print(f.type, f.location)
```

```
CDS features:
```

Task 8. Function that computes the reverse complement of a given DNA sequence.

Ensure that the input sequence is validated to contain only valid nucleotide characters (A, T, C, G).
Provide the output result for **the first 15 characters of the sequence.**



1) get first 15 characters
2) run reverse complement

Positive 5' A G T C A G T A A C T A G 3'

Negative 3' T C A G T C A T T G A T C 5'

Complementary to positive strand

5' C T A G T T A C T G A C T 3'

Reverse complementary to positive strand

Task 9. Function that takes DNA **(first 21 characters)** and performs both transcription and translation.

- Transcribe the DNA sequence into mRNA. In mRNA, thymine (T) is replaced by uracil (U).
- Translate the mRNA sequence into a protein sequence using the standard genetic code. **Do not forget about stop codons (they stop the translation).**

`Bio.Seq.transcribe`(*dna*)

Transcribe a DNA sequence into RNA.

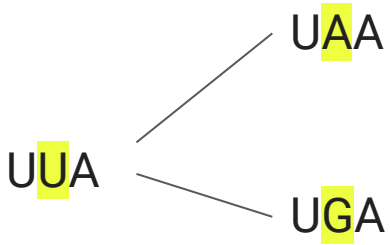
Following the usual convention, the sequence is interpreted as the coding strand of the DNA double helix, not the template strand. This means we can get the RNA sequence just by switching T to U.

```
DNA Sequence: GGTTAAACATGACTCCTAGAG
mRNA Sequence: GGUUAAACAUGACUCCUAGAG
Protein Sequence: G
```

```
DNA Sequence: GGTTAAACATGACTCCTAGAG
mRNA Sequence: GGUUAAACAUGACUCCUAGAG
Protein Sequence: G*T*LLE
```

Task 10. Write a function that calculates the total number of SNPs in all possible codons (except STOP codons) that can result in a stop codon (TAA, TAG, or TGA).

- Use Standard RNA codon table
- The simplest way to solve this task is to iterate over all possible codons and check how many single nucleotide changes (mutations) transform a sense codon into a stop codon.



TAA, TAG, or TGA (or UAA, UAG, UGA)

1st position SNP:

UAA:

- **C**AA
- **A**AA
- **G**AA

Result (9):

- ❑ AAA
- ❑ AGA
- ❑ AAG

UAG:

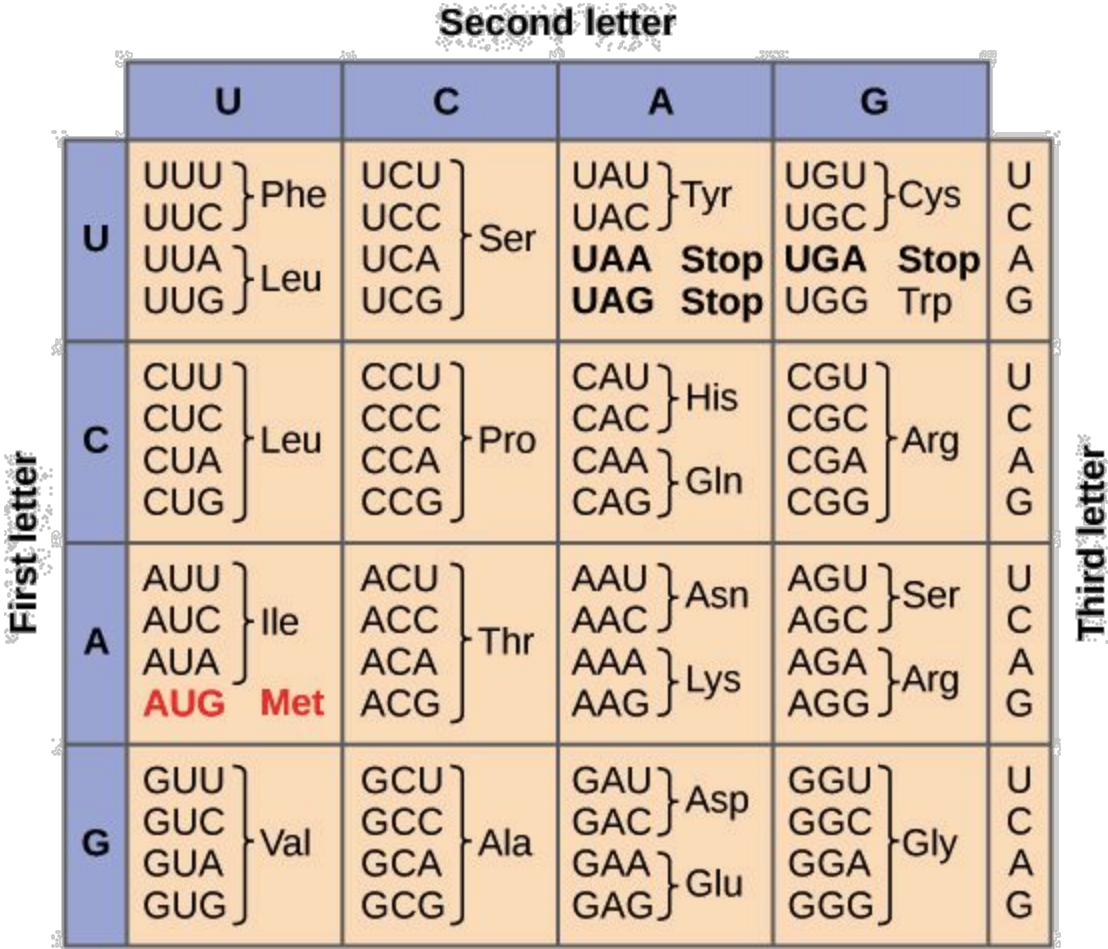
- **C**AG
- **A**AG
- **G**AG

- ❑ CAA
- ❑ CAG
- ❑ CGA

UGA:

- **C**GA
- **A**GA
- **G**GA

- ❑ GAA
- ❑ GAG
- ❑ GGA



TAA, TAG, or TGA (or UAA, UAG, UGA)

2nd position SNP:

UAA:

- U**U**A *
- U**C**A *
- U**G**A

Result (7):

- ❑ U**A**A ~~** STOP~~

UAG:

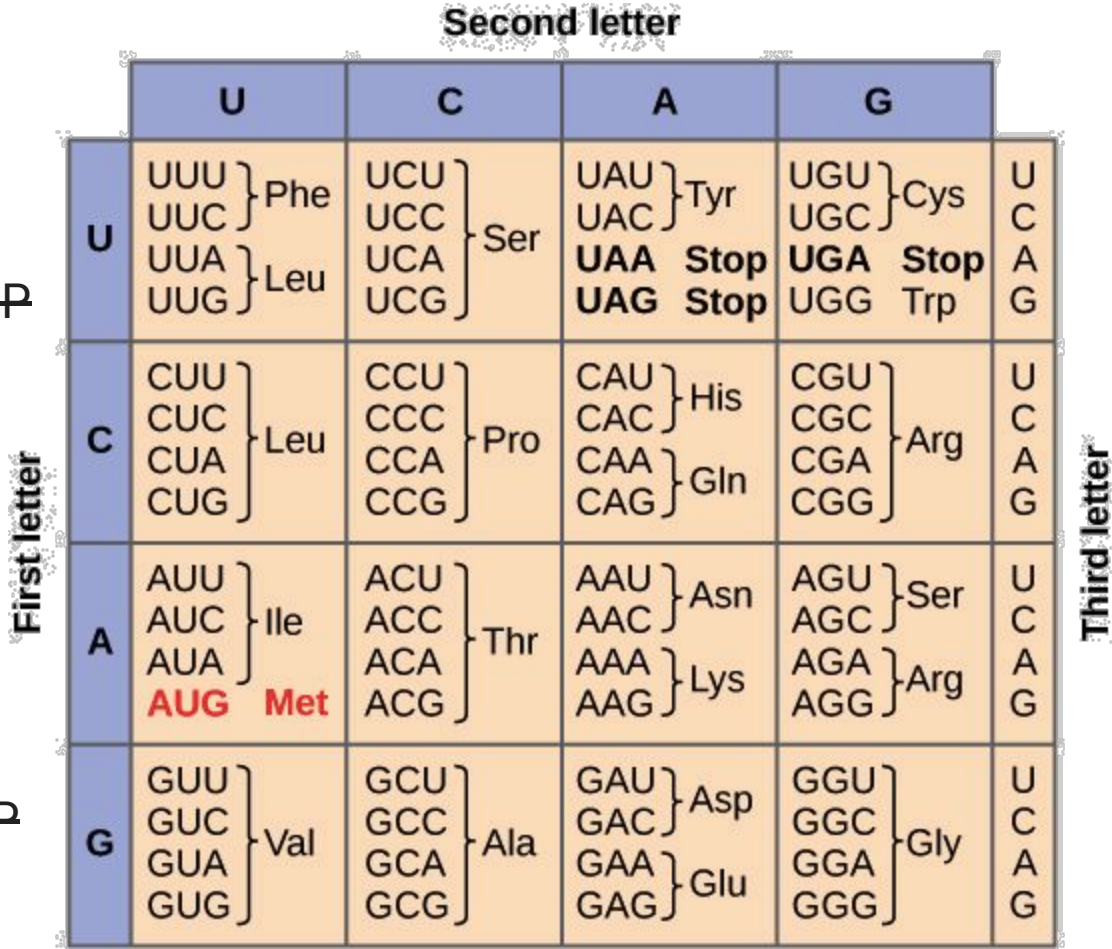
- U**U**G
- U**C**G
- U**G**G

- ❑ U**U**A x 2
- ❑ U**U**G
- ❑ U**C**A x 2
- ❑ U**C**G

UGA:

- U**U**A *
- U**CA** *
- U**A**A

- ❑ U**G**G
- ❑ U**G**A ~~** STOP~~



TAA, TAG, or TGA (or UAA, UAG, UGA)

3rd position SNP:

UAA:

- UA**U** *
- UA**C** *
- UA**G**

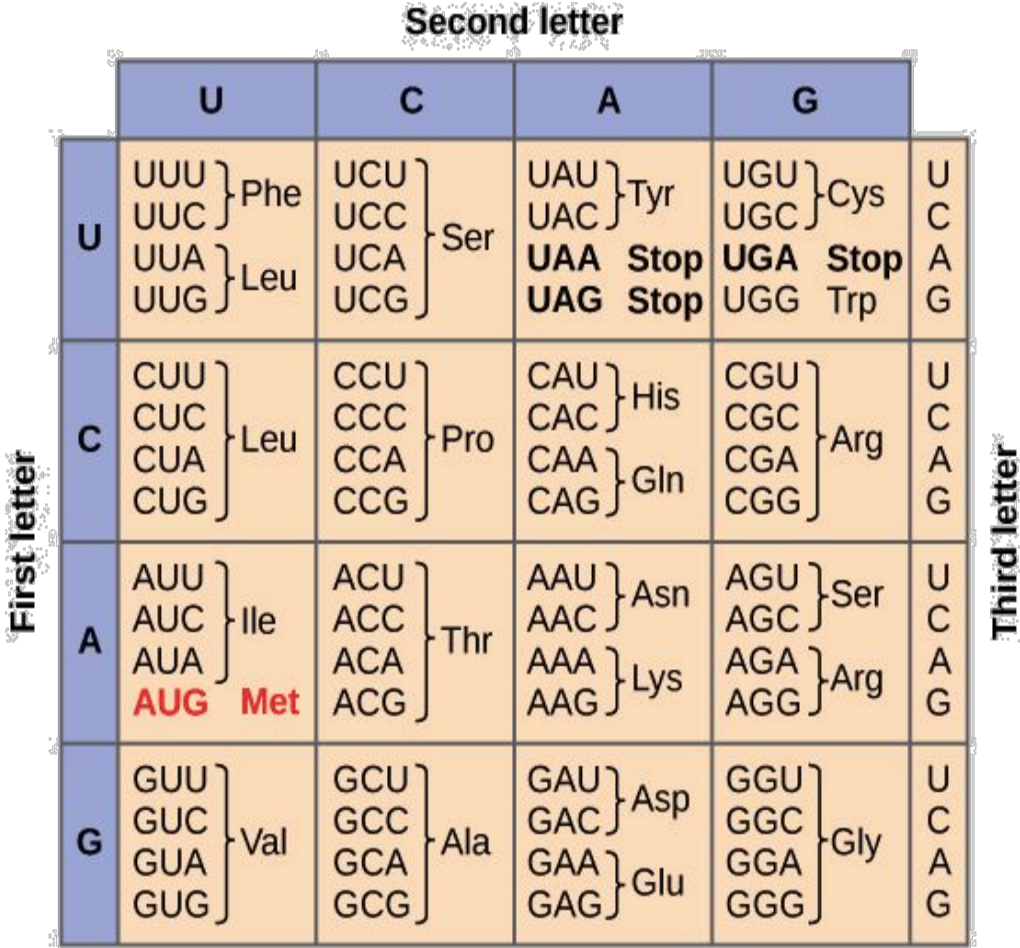
UAG:

- UA**U** *
- UA**C** *
- UA**A**

UGA:

- UG**U**
- UG**C**
- UG**G**

Result (7):

- ❑ UAA ~~** STOP~~
- ❑ UAU x 2
- ❑ UAC x 2
- ❑ UAG ~~** STOP~~
- ❑ UGU
- ❑ UGC
- ❑ UGG

9 + 7 + 7 = 23